

Final Audit Report

2026-05-15

Contents

Final Audit Report	1
Status distribution	1
Plan adherence	1
Confidence calibration	2
Residual debt	2
Findings by severity	2
Future work	2
Reconciliation log	2
Validator and coverage notes	2

Final Audit Report

Run id: run-2026-05-14T232311Z Document generated: 2026-05-15T13:26:26.155533+00:00

This final audit checked the run’s structured commitments, ledger states, cited evidence files, closure artifacts, and validator outputs. It did not re-prove the mathematics independently; it verified that the public record supports the run’s stated milestone outcomes and that computational checks are not presented as substitutes for proof.

Status distribution

- validated: 5

Plan adherence

Milestone	Terminal status	Confidence	Evidence count	Latest evidence pointer
M0	validated	high	2	docs/lemmas/lemma_catalogue.md
M1	validated	high	6	scripts/rr/finite_rr_experiment.py
M2	validated	high	12	docs/proof/bailey_matrix_transfer_mechanism.md
M3	validated	high	6	docs/proof/product_side_formalization.md
M4	validated	high	8	docs/proof/final_proof_synthesis_plus_validation_separation.md

All five plan milestones M0-M4 are latest-state **validated/high**. Stage verification confirmed that the cited evidence files exist and support the corresponding record: formal setup and validation separation for M0, finite experiment harness for M1, derived Bailey-style transfer mechanism for M2, product-side formalization for M3, and final proof synthesis plus validation separation for M4.

Key proof and validation documents present: - docs/lemmas/lemma_catalogue.md: present - docs/proof/bailey_matrix_transfer_mechanism.md: present - docs/proof/product_side_formalization.md: present - docs/proof/final_proof_synthesis_plus_validation_separation.md: present - docs/validation.md: present

Confidence calibration

- validated/high: 5
- validated/medium: 0
- validated/low: 0
- validated/provisional: 0
- Low-confidence terminal states: none

The high-confidence terminal states are supported by the ledger artifacts and by two adversarial test passes. The validation record keeps coefficient and residual checks in a discovery/regression role rather than using them as proof.

Residual debt

- `_run/start` (nonterminal-ledger-state): Latest ledger state is in-progress/high. This is bookkeeping debt rather than a proof defect for plan milestones M0-M4.

No residual debt affects the validated status of M0-M4. The remaining item is a run-state bookkeeping marker.

Findings by severity

- CRITICAL: 0
- MODERATE: 0
- MINOR: 0

No final-audit findings remain. Earlier verifier false positives were removed before this document stage because they came from audit-script parsing and evidence-classification mistakes, not from run defects.

Future work

- Anchored to `_run/start`: Close or supersede the run-start bookkeeping entry when the final audit artifacts are committed, so terminal reports do not retain an in-progress run marker.

Reconciliation log

- None. No `_plan/`, `_archive/`, or correction events are proposed by the final audit.

Validator and coverage notes

- `promise_check`: green
- `org_check`: green
- Stage files present: 5 / 5
- Figure coverage: 9 figure files present, 9 referenced by ledger, 0 missing, 0 orphaned.